

Dylan Smith Homework #1 Part 3

$$1A. \quad P(X=1) = \frac{37}{180} \quad P(X=2) = \frac{31}{180} \quad P(X=3) = \frac{14}{45} \quad P(X=4) = \frac{14}{45} \\ P(Y=0) = \frac{1}{3} \quad P(Y=1) = \frac{2}{5} \quad P(Y=2) = \frac{4}{15}$$

B. X and Y are not independent because $P(X=x, Y=y) \neq P_X(x) \cdot P_Y(y)$.
For example $P(X=1, Y=0) = \frac{1}{18}$ but $P_X(1) \cdot P_Y(0) = \frac{37}{180} \cdot \frac{1}{3} = \frac{37}{540}$
 $\frac{1}{18} \neq \frac{37}{540}$ so X and Y are not independent.

$$C. \quad P(Y=2) = \frac{4}{15} \quad P(X=1|Y=2) = \frac{1/15}{4/15} = \frac{1}{4} \quad P(X=2|Y=2) = \frac{1/30}{4/15} = \frac{1}{8} \\ P(X=3|Y=2) = \frac{1/30}{4/15} = \frac{1}{8} \quad P(X=4|Y=2) = \frac{2/15}{4/15} = \frac{1}{2} \\ E[X|Y=2] = 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{8} + 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{2} = \frac{23}{8} = 2.875$$

$$D) \quad E[Y] = \frac{14}{15} \quad E[XY] = \frac{51}{20} = 2.55$$

2. A. A and B are independent. The outcome of the first toss does not affect the second toss. None of the other pairs are independent.